

# Senior Design Project: Automatic Sorting System

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## Introduction

- **Goal:** Develop a multi-purpose recycling bin capable of automatically sorting waste into four categories:
  - Plastic
  - Metal
  - Glass
  - Organics (for items that don't fit the other categories or are mistakenly placed in the bin)
- **Purpose:** Simplify and automate the recycling process to:
  - Increase convenience
  - Improve the accuracy of waste sorting

## Methods

This project is centered around the capabilities of an AI-assisted camera, designed to recognize various objects and classify them into predefined categories. The system utilizes the **Huskylens AI camera** for object detection and classification.

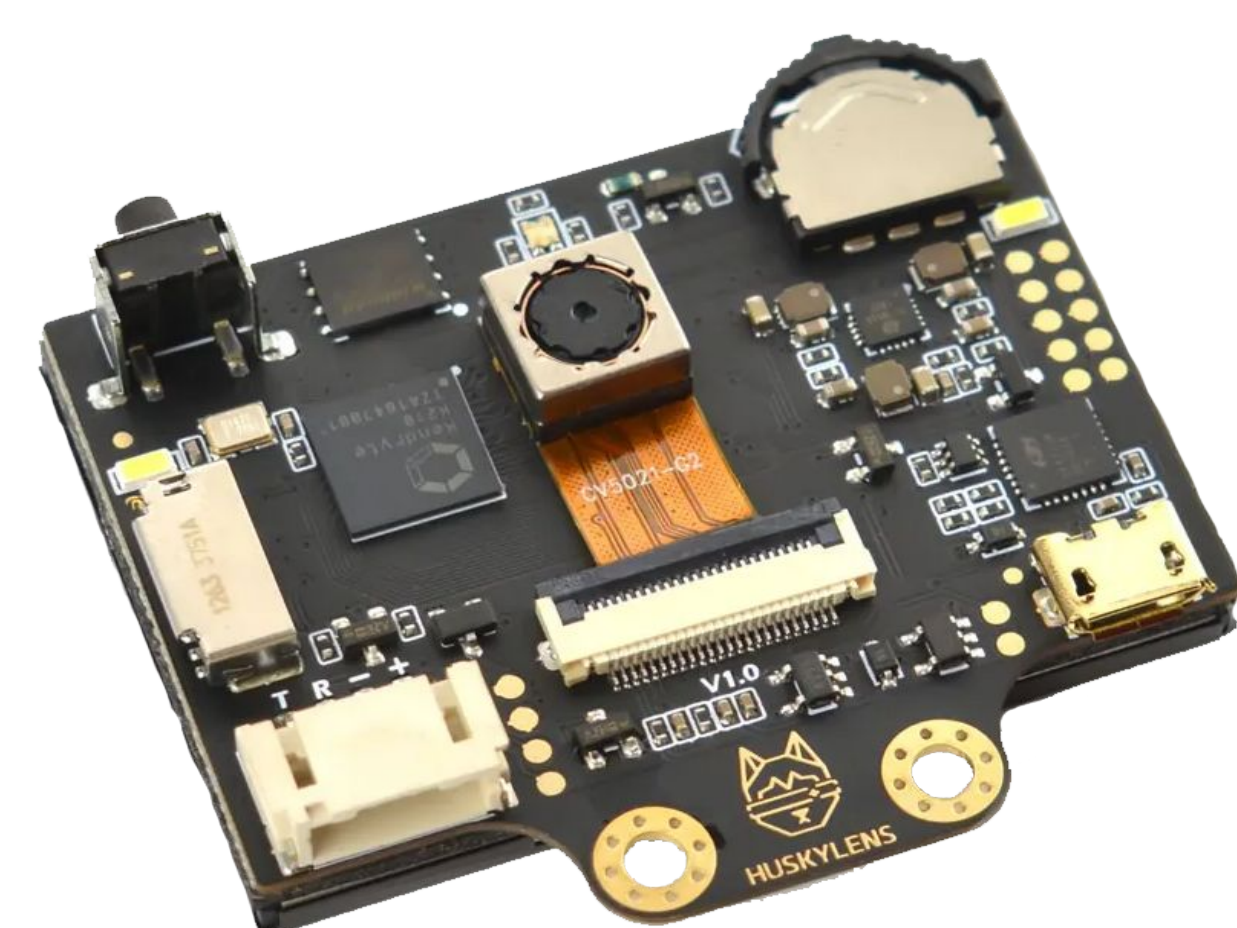


Figure 1: Huskylens Camera

The project methodology is divided into the following steps:

1. Research & Design
2. Hardware Assembly
3. Programming
4. Testing & Optimization

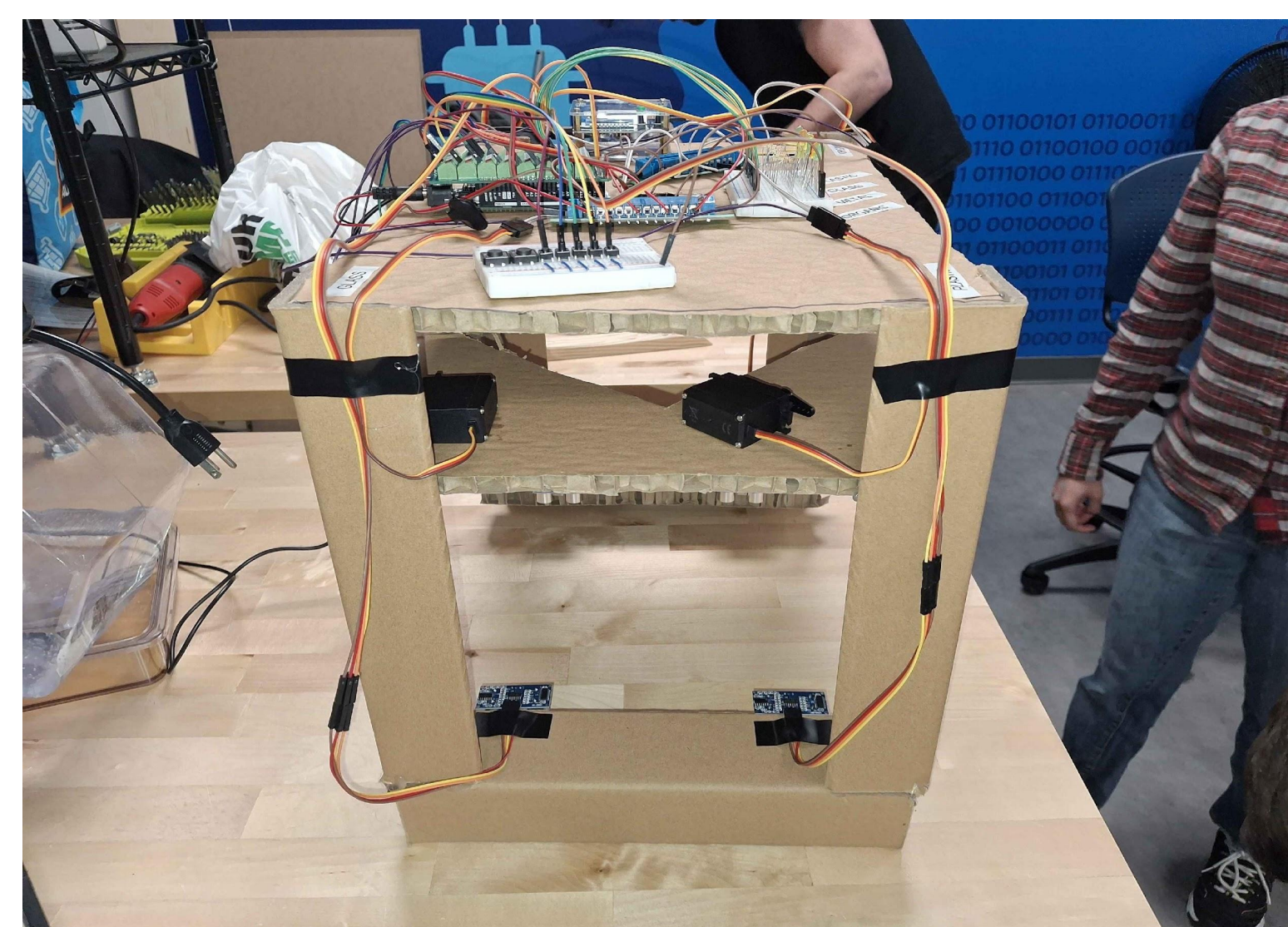


Figure 2: Electronics on a temporary setup

The electronics were initially put onto the cardboard frame, imitating the final product for testing purposes.

## Results

- **System Architecture:** The system utilizes a camera connected to an Arduino board, which serves as the central processing unit. The Arduino interfaces with the necessary electronic components.

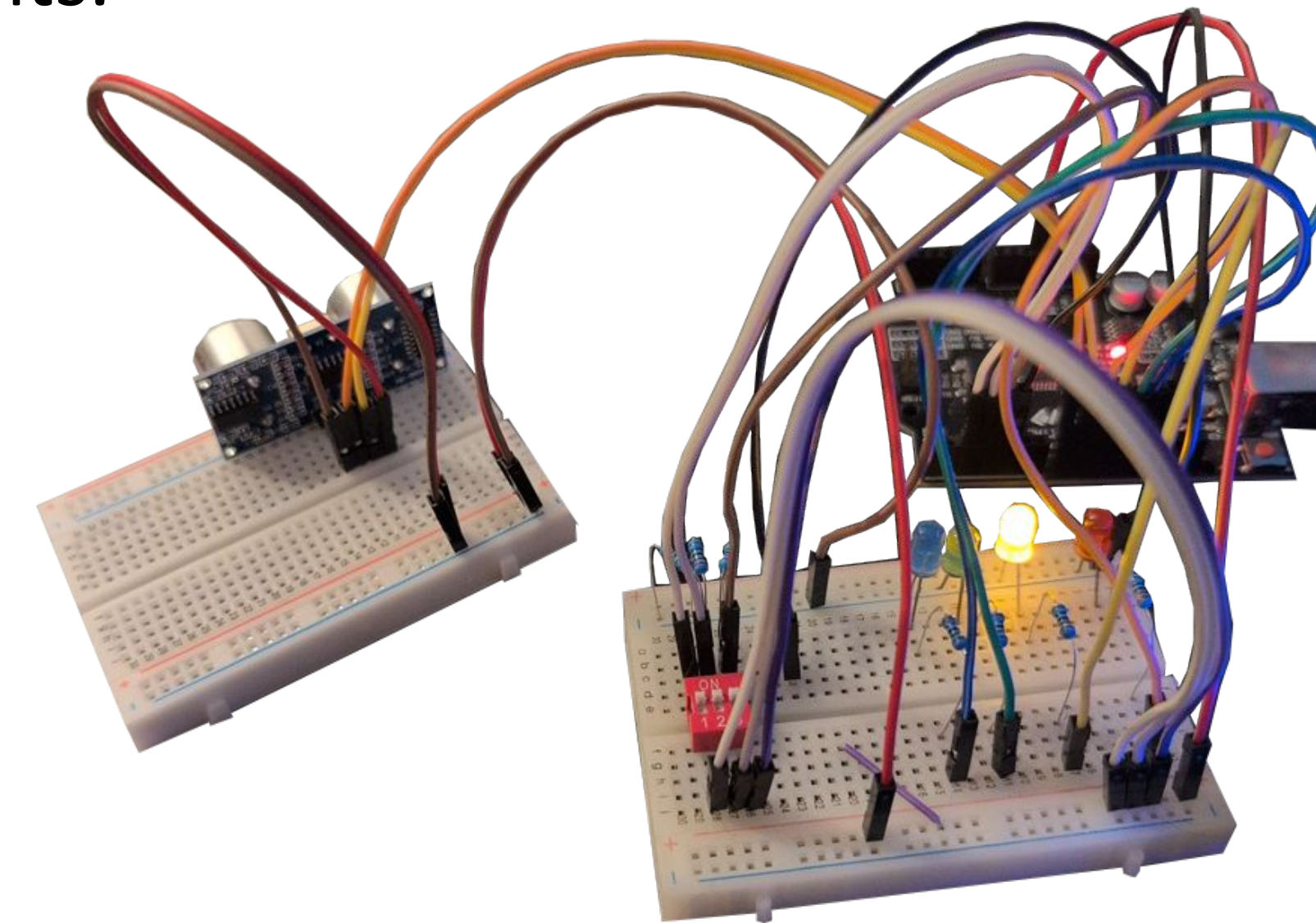


Figure 3: The circuit system for the controller

- **Bin Level Monitoring:** The system uses sonic sensors to monitor the fill level of each bin, indicating when they need to be emptied.
- **Automated Doors:** Doors operate automatically when an object is detected, with each door corresponding to a specific waste category (plastic, metal, glass, or garbage).
- **Sorting Process:** After the bin is assembled and the camera is trained to recognize objects, the system sorts the objects into the appropriate bins.

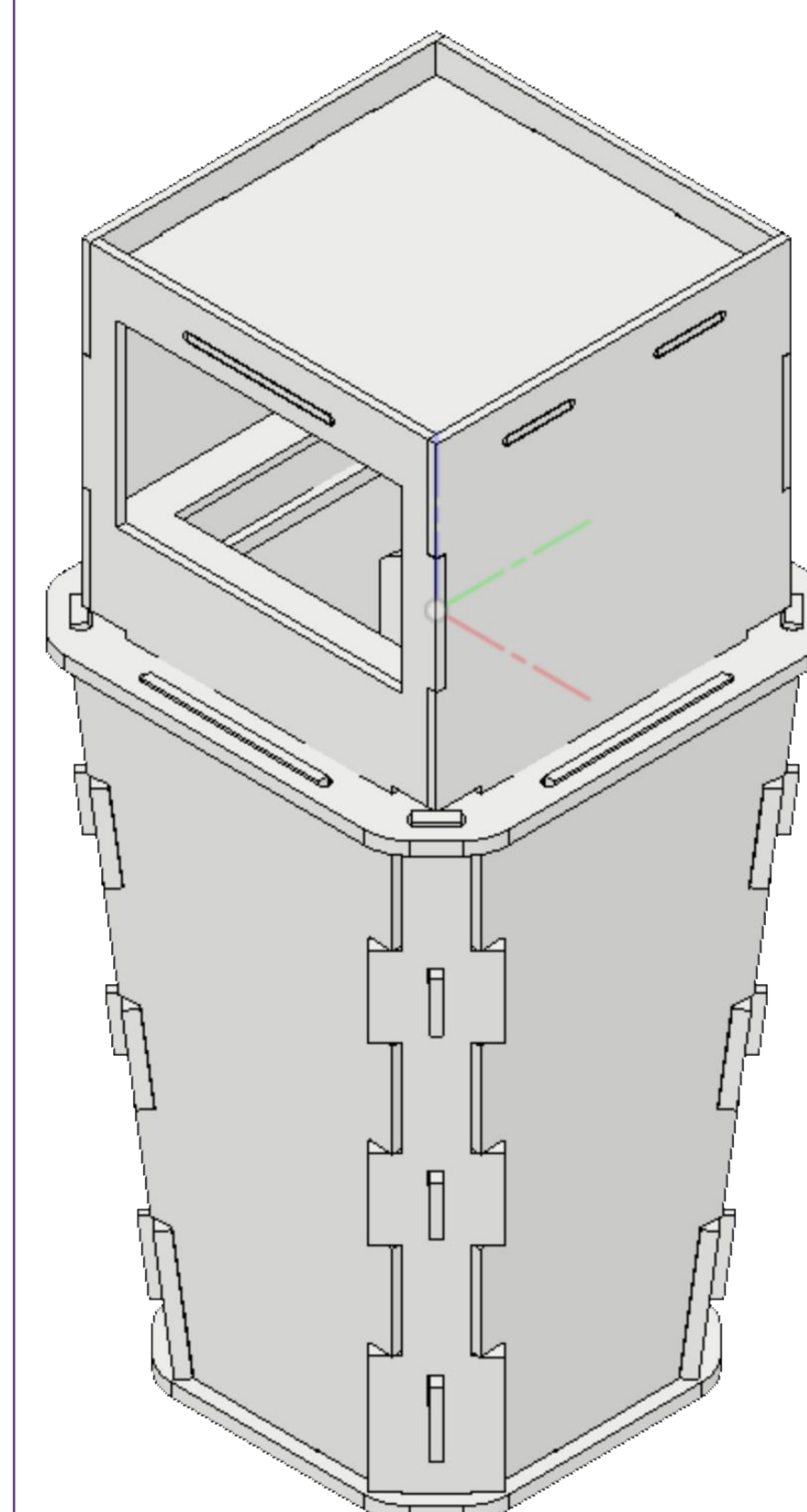


Figure 4: Render for the bin, containing the system

Top bin: houses all of the electronics

Bottom bin: contains all the sorted garbage and recyclables

|                                    |          |
|------------------------------------|----------|
| Arduino board:                     | \$43.56  |
| Electronics (Wires and motors):    | \$106.8  |
| Raw materials (Wood and hardware): | \$100    |
| Peripherals:                       | \$41.31  |
| Total:                             | \$291.67 |

Figure 5: Cost table

## Results Cont.

After an extended period of training the camera, it can now distinguish between 4 categories of objects:

- Metal
- Plastic
- Glass
- Trash

And as a result, the final components are being put into a more permanent, wooden frame.



Figure 6: The sorting gates

## Conclusion

- **Project Aim:** Provide a convenient, efficient, and user-friendly solution for recycling waste.
- **Future Development:**
  - Improve object recognition to handle a wider variety of items.
  - Potentially integrate a barcode reading system for increased accuracy.

## References:

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2. Adding AI Vision To A Robot Car Using A Huskylens - The DIY Life
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4. LIZARD: Pervasive Sensing for Autonomous Plastic Litter Monitoring | IEEE Conference Publication | IEEE Xplore
5. WasteMiner: An Efficient Waste Collection System for Smart Cities Leveraging IoT and Data Mining Technique - Experts@Syracuse

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